Department of Electronic & Telecommunication Engineering
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EN4431 - Analog IC Design

Assignment 03
Blockwise Layout Design of an Integer-N Phase Locked Loop

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1 Introduction

The objective of Assignment 3 was to continue the Integer-N Phase-Locked Loop design from the schematic level to the physical layout level using the SKY130 process. The main focus of this assignment was the blockwise layout implementation of the PLL, followed by physical verification using Design Rule Check (DRC) and Layout Versus Schematic (LVS).

2 Updated Team Distribution

During this stage, the team distribution was slightly changed from the initial grouping. The final block allocation was as follows:

Table 1: Updated team distribution

Team	**Assigned Block**	**Members**
Team 1	Phase Frequency Detector	Sajitha, Jazoole
Team 2	Charge Pump and VCO	Imasha, Charles
Team 3	Divider	Lohan, Erandee

I worked in Team 2, which was responsible for the Charge Pump and VCO layout design. My part was to complete the layout implementation of both the Charge Pump and the VCO until the DRC-clean stage. The LVS verification was done by Charles.

3 Individual Work Contribution

My individual work mainly covered the physical layout design of the Charge Pump and the VCO. The layout work included device placement, routing, power connection, signal routing, and DRC cleanup.

The main tasks I completed were:

1. Created the layout for the Charge Pump block.
2. Created the layout for the current-starved ring VCO block.
3. Routed the main signal, bias, power, and ground connections.
4. Checked the layouts using DRC.
5. Removed DRC violations until both layouts reached a DRC-clean state.
6. Prepared the blocks for LVS checking and team-level integration.

4 Charge Pump Layout Design

The Charge Pump is one of the most important analog blocks in the PLL because it converts the UP and DOWN signals from the PFD into current pulses. In the layout, a very simple approach was used to reduce the chance of failure.

The Charge Pump layout was completed with the required devices, interconnections, and supply routing. The layout was checked using DRC, and the rule violations were fixed until a DRC-clean layout was obtained.

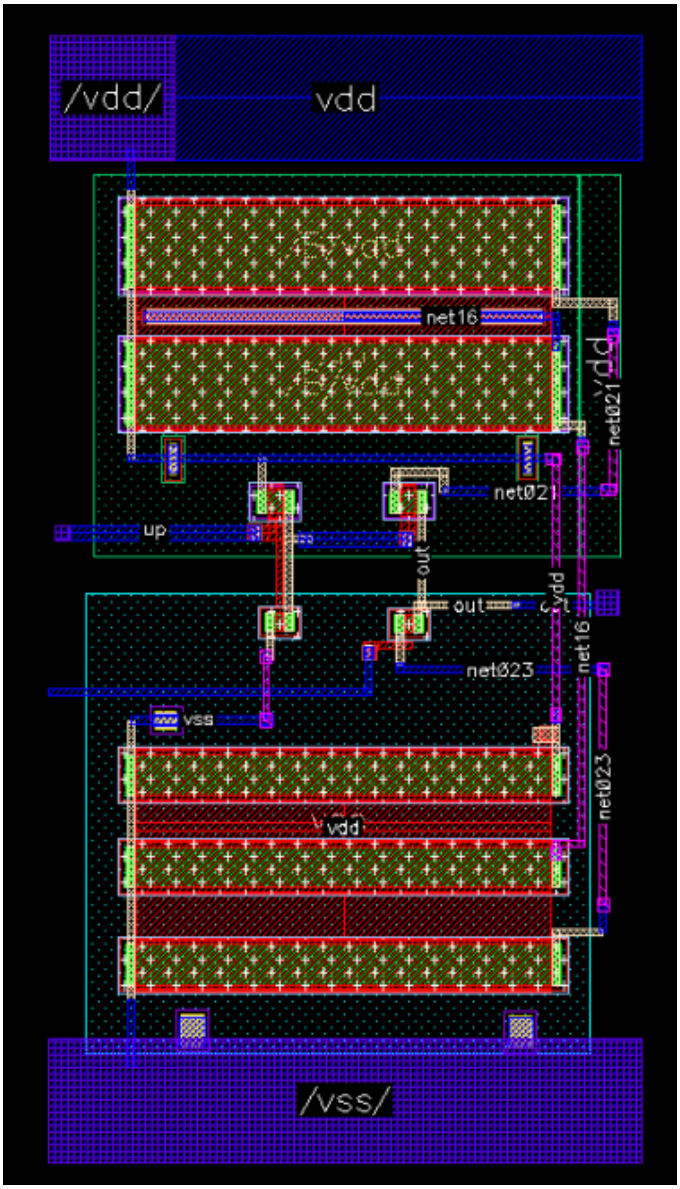


Figure 1: Layout of the charge pump

```
Total CPU Time                  : 0(s)
Total Real Time                 : 0(s)
Peak Memory Used                : 18(M)
Total Original Geometry         : 302(467)
Total DRC RuleChecks            : 578
Total DRC Results               : 0 (0)
Summary can be found in file Charge_Pump.sum
ASCII report database is /home/user19/PLL_421E/drc/Charge_Pump.drc_errors.ascii
Checking in all SoftShare licenses.
```

```
ERC Summary File            :   result.sum
Extraction Report File      :   Charge_Pump.rep
Comparison Report File      :   Charge_Pump.rep.cls
Top Cell                    :   Charge_Pump  <vs>  Charge_Pump

Run Result                  :   MATCH

Run Summary                 :   [INFO]  ERC Results: Empty
                            :   [INFO]  Extraction Clean
```

Figure 2: DRC and LVS results

5 VCO Layout Design

The VCO used in this PLL is a current-starved ring oscillator. In this topology, the oscillation frequency is controlled by changing the current available to the inverter delay stages. Therefore, the control voltage from the Charge Pump and loop filter changes the delay of the ring oscillator stages and adjusts the output frequency.

The VCO layout was implemented by placing the delay stages and biasing/current-starving devices in a compact symmetric arrangement. The main focus was to maintain proper connectivity, reduce unnecessary routing length, and keep the layout suitable for physical verification.

The VCO layout was also checked using DRC. All DRC violations found during layout verification were corrected, and the final VCO layout was made DRC clean.

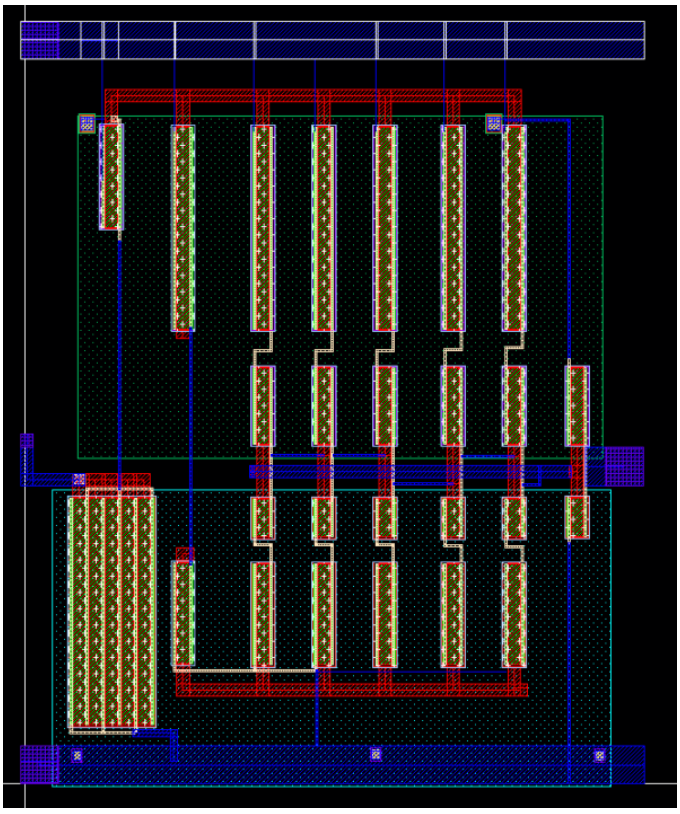


Figure 3: Layotu of the VCO

```
Total CPU Time                  : 0(s)
Total Real Time                 : 0(s)
Peak Memory Used                : 18(M)
Total Original Geometry         : 966(2308)
Total DRC RuleChecks            : 578
Total DRC Results               : 0 (0)
Summary can be found in file Current_trarved_ring_VCO.sum
ASCII report database is /home/user19/PLL_421E/drc/Current_trarved_ring_VCO.drc_errors.ascii
Checking in all SoftShare licenses.
```

```
############################################################
#                                                          #
# Run Result             : MATCH                           #
#                                                          #
# Run Summary            : [INFO]  ERC Results: Empty      #
#                          : [INFO]  Extraction Clean      #
#                                                          #
# ERC Summary File       : result.sum                      #
# Extraction Report File : Current_trarved_ring_VCO.rep    #
# Comparison Report File : Current_trarved_ring_VCO.rep.cls #
#                                                          #
############################################################
```

Figure 4: DRC and LVS results

6 Issues Faced During Layout

One of the main issues faced during the layout work was related to adding guard rings. Initially, I expected guard rings to behave like a separate layout entity, similar to how they can be handled in Magic. However, in the Cadence/SKY130 layout environment, guard rings do not seem to be supported as a separate entity in this PDK, and it needs to be studied further.

When I attempted to add guard rings, errors appeared in the layout verification stage for same working PDK with pcell evaluation failure. Because of this, the final A3 submission focused on making the Charge Pump and VCO layouts DRC clean first. The guard-ring implementation is planned as an improvement for the final tapeout submission.

7 Improvements Planned for Final Tapeout

Although the Charge Pump and VCO layouts were completed up to DRC-clean status, several improvements are planned before the final September tapeout submission.

7.1 Current Mirror Matching Improvement

The Charge Pump and VCO performance strongly depends on the matching between current mirror devices. In the current layout, further improvement can be made by using better analog matching techniques. For the final version, the current mirror layout should be improved using:

- common-centroid or interdigitated placement,
- identical transistor orientation (already follwoed),
- equal routing length for matched devices,
- dummy devices at the edges,
- symmetric source, drain, and gate routing,
- matched via and contact placement.

These improvements will reduce current mismatch between the UP and DOWN current paths and staerved currents in VCO. This is important because Charge Pump mismatch can introduce control-voltage ripple, static phase offset, and reference spur components in the PLL output, and also an unstable VCO would lead to unexpected locking frequencies.

7.2 Better Placement to Reduce Area

Another improvement is to optimize the device placement to reduce the total layout area. Since the target design should be compact for Tiny Tapeout integration, unused spaces and long routing paths should be reduced.

For the final tapeout version, the layout can be improved by:

- placing related devices closer together following minimum distance rules,
- folding large transistors where suitable,
- sharing diffusion where possible,
- aligning current mirror devices symmetrically,
- arranging VCO delay stages in a more compact structure,
- reducing unnecessary routing bends and detours,
- using higher metal layers for cleaner power and signal routing where allowed without the metal 5.

This will reduce the overall area and improve layout regularity.

7.3 Guard Ring Addition for Signal Integrity

Guard rings should be added around sensitive analog blocks, especially the VCO and Charge Pump. The VCO is sensitive to supply and substrate noise because noise at the control node or substrate can modulate the oscillation frequency and increase jitter or phase noise.

For the final tapeout version, guard rings will be implemented correctly using SKY130-compatible substrate and well tap structures around separate NMOS and PMOS blocks and Digital blocks. The planned approach is:

- add p-substrate guard rings connected to VSS around NMOS-sensitive regions,
- add n-well guard rings connected to VDD around PMOS regions,
- place guard rings around the VCO to reduce substrate noise coupling,
- keep noisy digital routes away from the VCO control node.

This improvement is expected to enhance signal integrity, reduce substrate coupling, and improve the robustness of the PLL layout.

7.4 Post-Layout Simulation After PEX

For A3, post-layout simulation after parasitic extraction was not mandatory. However, for the final tapeout submission, PEX should be performed after LVS completion. This will allow the extracted parasitic capacitances and resistances to be included in simulation.

These checks are important before final GDS submission because layout parasitics can change the performance compared to schematic-level simulations.

8 Final Tapeout Plan

For the final September tapeout submission, the main focus will be to refine the existing DRC-clean layouts and complete the full verification flow. The planned final steps are:

1. Improve Charge Pump current mirror matching.
2. Compact the placement to reduce the layout area.
3. Correctly add guard rings using SKY130-compatible structures.
4. Perform PEX and post-layout simulations.